

Avengers Assembly: We Didn't Know What You'd Like, So We Put Everything In

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Abstract. We present the non-standard cosmology suite of the cuRAMSES adaptive mesh refinement (AMR) code (cuRAMSES III), which assembles, within a single code base, essentially every popular extension of Λ CDM: (i) **self-interacting dark matter** (SIDM) with constant, velocity-dependent, anisotropic and inelastic (two-state) cross-sections, implemented as cell-based Monte Carlo scattering with machine-precision momentum and energy conservation; (ii) a **dark energy and non-GR gravity** battery comprising CPL dark energy with clustering perturbations, massive-neutrino linear response, early dark energy, field-level quintessence and purely kinetic k-essence, coupled quintessence, parametrized quasi-static Horndeski gravity, Hu–Sawicki $f(R)$, nDGP, symmetron, the environmentally damped dilaton, the parameter-free tracker cubic Galileon, and QUMOND — all built on a unified Poisson-solver framework in which model corrections compose multiplicatively, with an FFT-accelerated Newton stage for the nonlinear scalar fields; (iii) **fuzzy dark matter** (FDM), a hybrid Schrödinger–Poisson integrator with a spectral kinetic operator on the base level and finite differences on refined levels; and (iv) **evaporating primordial black holes** (PBH), a table-driven global mass-function scheme with optional local heating. Every dark-energy and gravity module has been verified at the equation level against the literature, at the solver level against analytic linear responses (to 10^{-5} or better), and end-to-end with a controlled 64^3 simulation suite compared against an initial-conditions-consistent linear forward model, agreeing to 2–5 per cent with residual signs explained by screening. We document the algorithms, unit conventions, namelist interfaces and the verification methodology, providing a self-contained reference for non-standard cosmological simulations with AMR codes.

Keywords: cosmological simulations, dark energy theory, modified gravity, dark matter theory

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Contents

1	Introduction	1
2	Self-Interacting Dark Matter	2
2.1	Physical motivation	2
2.2	Monte Carlo scattering algorithm	2
2.3	Cross-sections and angular distributions	2
2.4	Inelastic (two-state) SIDM	3
2.5	Timestep, threading and diagnostics	3
3	Dark Energy and non-GR Gravity	3
3.1	Unified Poisson-solver framework	3
3.2	CPL dark energy with clustering perturbations	4
3.3	Massive neutrino linear response	4
3.4	Early dark energy	4
3.5	Scalar-field dark energy: quintessence and k-essence	4
3.6	Coupled quintessence	5
3.7	Parametrized Horndeski gravity	5
3.8	Screened modified gravity: the scalar-field solvers	5
3.8.1	$f(R)$ Hu–Sawicki	5
3.8.2	nDGP	5
3.8.3	Symmetron	5
3.8.4	Environmentally damped dilaton	6
3.8.5	Tracker cubic Galileon	6
3.8.6	QUMOND	6
3.9	Numerical solution of the scalar equations	6
3.10	Verification	7
4	Fuzzy Dark Matter	7
5	Evaporating Primordial Black Holes	8
6	Conclusions	8

1 Introduction

The Λ CDM concordance model provides an excellent fit to a wide range of cosmological observations [30], yet the physical nature of both dark components remains unknown. On the dark-energy side, the cosmological constant may be replaced by a dynamical field with an evolving equation of state [9, 23], by an early dark energy component [31], or the acceleration may signal a breakdown of General Relativity on cosmological scales, as in $f(R)$ gravity [14], braneworld models [37], Galileon [4], symmetron and dilaton theories [5, 13], or MOND [27]. On the dark-matter side, small-scale tensions motivate self-interacting dark matter [38, 41], ultralight (fuzzy) dark matter [15, 36], and primordial black holes [8], while neutrino oscillations guarantee a small but non-zero hot component [19].

Stage-IV surveys will constrain all of these at the per-cent level, and their nonlinear signatures can only be predicted by simulations. Historically each extension has lived in its own specialised code; comparing models then convolves physical differences with code systematics. This paper describes the opposite approach: one AMR code, `cuRAMSES` [18], carrying *all* of the above extensions behind uniform namelist switches, with shared infrastructure (domain decomposition, Morton-key octree, multigrid/CG/FFT Poisson solvers, OpenMP+MPI parallelism) and a common verification methodology. Hence the title¹.

The paper is organised in four physics sections, one per family: Section 2 describes self-interacting dark matter; Section 3 the dark energy and non-GR gravity battery, including the unified Poisson framework, the nonlinear scalar-field solvers and their verification; Section 4 fuzzy dark matter; and Section 5 evaporating primordial black holes. Section 6 concludes. Throughout, code quantities use the supercomoving variables of Martel & Shapiro [26]: $\tilde{x} = x_{\text{com}}/(aB)$ (with B the box size), $\tilde{\rho} = \rho a^3/(\rho_c \Omega_m)$ (so $\tilde{\rho} = 1$), $\tilde{\phi} = a^2 \phi/(BH_0)^2$ and $d\tilde{t} = H_0 dt/a^2$.

2 Self-Interacting Dark Matter

2.1 Physical motivation

SIDM models introduce a non-zero scattering cross-section σ/m that allows dark matter particles to exchange momentum and energy, potentially resolving the core-cusp, missing satellites and too-big-to-fail problems [38, 41]. Astrophysical constraints span dwarf galaxies ($v \sim 30\text{--}100 \text{ km s}^{-1}$) to clusters ($v \sim 10^3\text{--}3 \times 10^3 \text{ km s}^{-1}$); a constant $\sigma/m \sim 1 \text{ cm}^2 \text{ g}^{-1}$ fits dwarf scales but may violate cluster bounds, motivating velocity-dependent cross-sections from light-mediator models [11, 17, 42].

2.2 Monte Carlo scattering algorithm

We implement cell-based Monte Carlo scattering following the standard mesh/SPH approach [33, 34]. In every AMR leaf cell, dark matter particles are collected from the particle linked list, shuffled with the Fisher–Yates algorithm, and paired sequentially. For a pair with relative velocity v_{rel} the scattering probability per step is

$$P = \frac{\sigma/m(v) \bar{m} v_{\text{rel}} \Delta t}{V_{\text{cell}}} (N_{\text{dm}} - 1), \quad (2.1)$$

with $\bar{m}$ the mean particle mass, V_{cell} the physical cell volume and the factor $(N_{\text{dm}} - 1)$ accounting for all possible partners. A scatter occurs when a uniform deviate falls below P ; post-scattering velocities are constructed in the centre-of-mass frame,

$$\mathbf{v}'_i = \mathbf{v}_{\text{cm}} + \frac{m_j}{m_i + m_j} |v'_{\text{rel}}| \hat{\mathbf{n}}, \quad \mathbf{v}'_j = \mathbf{v}_{\text{cm}} - \frac{m_i}{m_i + m_j} |v'_{\text{rel}}| \hat{\mathbf{n}}, \quad (2.2)$$

which conserves momentum by construction and kinetic energy exactly for elastic scattering.

2.3 Cross-sections and angular distributions

Three velocity dependences are supported (`sidm_type`): constant $\sigma/m = \sigma_0$; Yukawa (Born-regime) $\sigma(v)/m = \sigma_0[1 + (v/v_0)^2]^{-2}$; and a generic power law $\sigma(v)/m = \sigma_0(v/v_0)^n$. The angular distribution is isotropic by default, or forward-peaked Rutherford-like, $d\sigma/d\Omega \propto (1 - \cos \theta + 2\varepsilon)^{-2}$ [16], sampled by CDF inversion.

¹From the Korean “*mwol joahalji mollaseo da neoesseo*” — we didn’t know what you’d like, so we put everything in.

Table 1. The dark energy and gravity battery. “Linear response” lists the analytically validated unscreened limit; the last column the accuracy achieved in dedicated unit tests of the discretised equations (see Section 3.10).

Model	Switch / namelist	Background	Linear response	Unit-test accuracy
CPL DE + perturbations	<code>w0,wa,cs2.de,de.table</code>	$w = w_0 + w_a(1 - a)$	Sapone–Kunz closure	$\leq 0.1\text{--}0.8\%$ vs CAMB ($z \geq 0.5$)
Massive ν linear response	<code>use_neutrino</code>	—	$1 + (\Omega_\nu/\Omega_{cb})R_\nu(k, a)$	table-exact
Early dark energy	<code>use_ede</code>	Poulin+19 fluid	smooth	analytic dilution exact
Quintessence	<code>use_quintessence</code>	Klein–Gordon (shooting)	smooth ($c_s^2 = 1$)	w -consistency 2×10^{-5}
k-essence	<code>use_kessence</code>	$P(X) = -X + X^2$ (exact)	Helmholtz with $c_s^2(a)$	analytic limits exact
Coupled quintessence	<code>use_coupled.de</code>	KG + mass evolution	$G_{\text{eff}} = G(1 + 2\beta^2)$	isolation runs $\leq 1\%$
Horndeski $\mu(a, k)$	<code>use_horndeski</code>	Λ CDM/CPL	$\mu = 1 + \mu_0\Omega_{\text{DE}}(a)/\Omega_{\text{DE},0}$	exact by construction
$f(R)$ Hu–Sawicki	<code>use_fR</code>	Λ CDM	$\mu = 1 + \frac{1}{3}\frac{k^2}{k^2 + m^2 a^2}$	6×10^{-7}
nDGP	<code>use_nDGP</code>	Λ CDM	$G_{\text{eff}}/G = 1 + 1/(3\beta)$	force field 1.5×10^{-6}
Symmetron	<code>use_symmetron</code>	Λ CDM	$F_5/F_N = 2\beta^2\chi^2$	exact (Brax+12 eq. 68 match)
Dilaton (Brax+12)	<code>use_dilaton</code>	Λ CDM	$2\beta(a)^2 k^2/(k^2 + m^2 a^2)$	5 digits
Cubic Galileon (tracker)	<code>use_galileon</code>	tracker $E^2(a)$	$G_{\text{eff}}/G(a=1) = 1.844$	5 digits
QUMOND	<code>use_mond</code>	—	$\nu(g_N /a_0)$	$\mu\nu$ conjugacy 2×10^{-16}

2.4 Inelastic (two-state) SIDM

A two-state model [40, 43] adds a mass splitting δ : ground–ground pairs can up-scatter ($g + g \rightarrow e + e$, allowed when $\frac{1}{2}\mu v_{\text{rel}}^2 > 2\delta$), excited pairs down-scatter exothermically, and mixed pairs scatter elastically; the relative speed is adjusted to conserve total (kinetic + internal) energy. An initial excited fraction is assigned at start-up in a restart-safe manner. A dissipative channel (`sidm_fdiss`) and a DM–baryon drag term (`sidm_baryon`) are also available.

2.5 Timestep, threading and diagnostics

The maximum per-level probability P_{max} is fed back into the timestep as $\Delta t_{\text{new}} \leq (C_{\text{SIDM}}/P_{\text{max}}) \Delta t_{\text{old}}$ (default $C_{\text{SIDM}} = 0.1$). The sweep is OpenMP-parallel over grids with per-thread decorrelated RNG streams and deterministic reseeding, and cumulative momentum/energy residuals are reduced across MPI ranks each step, providing runtime verification of the conservation properties. All parameters live in `&SIDM_PARAMS`.

3 Dark Energy and non-GR Gravity

This section describes the largest part of the suite: eleven dark-energy and modified-gravity models sharing one solver infrastructure. Table 1 summarises the models, their parameters and their verification status.

3.1 Unified Poisson-solver framework

In supercomoving units the code solves $\tilde{\nabla}^2 \tilde{\phi} = \frac{3}{2}\Omega_m a(\tilde{\rho} - 1)$. All smooth-dark-sector corrections enter this equation as multiplicative factors on the source. A single function, `cosmo_poisson_fourpi()`, provides the prefactor consistently to the multigrid, conjugate-gradient and coarse-level solvers, and the FFT solvers apply the same corrections per Fourier mode in the Green’s function, so that scale-dependent responses cost nothing extra. The factors compose:

$$\frac{3}{2}\Omega_{cb} a \times \underbrace{\left[1 + \frac{\Omega_\nu}{\Omega_{cb}} R_\nu(k, a)\right]}_{\text{neutrinos}} \underbrace{\left[1 + \frac{\Omega_{\text{DE}}(a)}{\Omega_{cb}} R_{\text{DE}}(k, a)\right]}_{\text{DE clustering}} \underbrace{\mu(a, k)}_{\text{Horndeski coupled DE}} \underbrace{e^{-\beta(\varphi - \varphi_0)}}_{\text{Horndeski coupled DE}}. \quad (3.1)$$

3.2 CPL dark energy with clustering perturbations

The background uses the exact CPL integral $f_{\text{DE}}(a) = a^{-3(1+w_0+w_a)}e^{-3w_a(1-a)}$ in the supercomoving Friedmann factors, the growth factor and the initial velocity scaling. Perturbations are supported in two modes:

(i) *Table mode.* A CAMB-derived table of the effective ratio $R_{\text{DE}}(k, a) = \delta_{\text{DE}}/\delta_m$ is measured with the Weyl-potential method (the ratio of potential per unit matter contrast between a clustering-DE and a smooth-DE universe), which yields the Newtonian-gauge response appropriate for the Poisson source; synchronous-gauge δ_{DE} would over-correct by an order of magnitude.

(ii) *Quasi-static fallback.* When no table is supplied and $c_s^2 > 0$, the code applies the Sapone & Kunz [35] closure

$$\frac{\delta_{\text{DE}}}{\delta_m} = \frac{1+w}{(1-3w) + \frac{2c_s^2 k^2}{3a^2 H^2}}, \quad \text{i.e. d.f.} = 1 + \frac{\alpha}{k^2 + \kappa^2}, \quad (3.2)$$

with $\kappa^2 = \frac{3}{2}(1-3w)a^2 E^2 (L/\lambda_H)^2 / c_s^2$ and $\alpha = (\Omega_\Lambda f_{\text{DE}} a^3 / \Omega_m)(1+w) \frac{3}{2} a^2 E^2 (L/\lambda_H)^2 / c_s^2$. Correct limits: a k -independent boost $1 + (\rho_{\text{DE}}/\rho_m)(1+w)/(1-3w)$ for $c_s^2 \rightarrow 0$, unity below the sound horizon, and bit-wise unity for Λ . Validated against CAMB (same Weyl method) to ≤ 0.1 per cent for $(w_0, w_a) = (-0.9, 0.1)$ and ≤ 0.8 per cent for $w_0 = -0.8$ at $z \geq 0.5$ (~ 4 per cent at $z = 0$, where the quasi-static approximation degrades — use the table mode for precision at low redshift). An earlier release used a closure with the opposite sign (a 15–29 per cent source suppression); results obtained with it should be discarded.

3.3 Massive neutrino linear response

Neutrino perturbations are added without neutrino particles: $\delta_\nu(k, a) = R_\nu(k, a) \delta_{cb}(k, a)$ with R_ν precomputed from CAMB [20], entering the Poisson source as in Section 3.1. This avoids neutrino shot noise entirely and is exact at linear order [cf. 1].

3.4 Early dark energy

The Poulin et al. [31] fluid form $\rho_{\text{EDE}}(a) = \omega_{\text{ede}} \rho_{\text{std}}(a_c) 2/[1 + (a/a_c)^{3(1+w_{\text{ede}})}]$ is added to the Friedmann factors. The implementation reproduces the analytic dilution slope $-3(1+w)$ and the analytic peak $f_{\text{EDE}}^{\text{max}} = 1.14 \omega_{\text{ede}}$ at $a = 3^{1/4} a_c$ exactly. Two conventions matter when mapping CMB constraints: ω_{ede} is the fraction at a_c (not the peak), and the background carries no radiation, so at $z_c \sim 3000$ the fraction differs from Planck-style f_{EDE} by $\sim 2\times$.

3.5 Scalar-field dark energy: quintessence and k-essence

A dedicated module tabulates scalar-field backgrounds on a $\ln a$ grid ($a = 10^{-6}$ – 1.55) with lazy, self-healing initialisation (the tables rebuild automatically if the cosmology is overridden by IC headers).

Quintessence [32]: the Klein–Gordon and Friedmann equations are integrated with RK4 for Ratra–Peebles ($V \propto \varphi^{-\alpha}$) or exponential potentials, the amplitude fixed by shooting to $\Omega_\varphi(a=1) = \Omega_\Lambda$. The tabulated $w(a)$ obeys $d \ln f_{\text{DE}} / d \ln a = -3(1+w)$ to 2×10^{-5} , and the RP tracker attractor $w = -2/(\alpha + 2)$ is recovered in the matter era.

k-essence [3]: for the purely kinetic $P(X) = -X + X^2$ the shift-symmetry integral $P_{,X} \sqrt{2X} a^3 = \text{const}$ gives $X(a)$ algebraically; $w \rightarrow 1/3$ and $c_s^2 \rightarrow 1/3$ at early times (a dark-radiation-like component) and $w \rightarrow -1$ today. The model’s $c_s^2(a)$ feeds the Helmholtz closure of equation (3.2) automatically.

3.6 Coupled quintessence

Conformally coupled dark matter, $m_{\text{dm}} \propto e^{-\beta\varphi/M_{\text{Pl}}}$ [2], adds four effects, all implemented and individually verified: (i) the modified expansion, including the DM mass-evolution factor in the *matter* term of the Friedmann integrals (its omission is detectable: an isolation run reproduced the inconsistent- H growth prediction to 0.2 per cent); (ii) the same factor multiplying the Poisson source; (iii) the momentum-conservation velocity term $+\beta\dot{\varphi}\mathbf{v}$, split across the two half-kicks of the KDK integrator; and (iv) the fifth-force enhancement $G_{\text{eff}} = G(1 + 2\beta^2)$ between matter particles.

3.7 Parametrized Horndeski gravity

The quasi-static, scale-independent limit of Horndeski gravity is implemented as $\mu(a) = 1 + \mu_0 \Omega_{\text{DE}}(a)/\Omega_{\text{DE},0}$ multiplying the Poisson source, with an optional Compton-mass k -dependence $\mu(a, k) = 1 + [\mu(a) - 1] k^2/(k^2 + a^2 m^2)$ applied exactly in the FFT solver paths (the multigrid path uses the $k \rightarrow \infty$ limit).

3.8 Screened modified gravity: the scalar-field solvers

Five models solve a nonlinear elliptic equation for an extra scalar degree of freedom on the AMR hierarchy, with the fifth force added to the particle acceleration.

3.8.1 $f(R)$ Hu–Sawicki

The quasi-static equation in code units is

$$\tilde{\nabla}^2 f_R = \frac{a^2}{3} \left(\frac{H_0 B}{c} \right)^2 [\tilde{R}(f_R) - \tilde{R}] - \Omega_m \left(\frac{H_0 B}{c} \right)^2 \frac{\tilde{\rho} - 1}{a}, \quad (3.3)$$

with the Hu–Sawicki inversion $\tilde{R} = \tilde{R}_0 (f_{R0}/f_R)^{1/(n+1)}$ and force $F_5 = +\frac{a^2}{2} (c/H_0 B)^2 \tilde{\nabla} f_R$ [14, 21, 29]. The linearized mass term is a Yukawa (the Jacobian is negative definite) and the chameleon limit $\tilde{R} \rightarrow \tilde{R} + 3\Omega_m \delta/a^3$ is exact. The discretised linear response reproduces $\mu(k) - 1 = \frac{1}{3} k^2/(k^2 + m^2 a^2)$ to 6×10^{-7} across $a \in [0.25, 1]$, $f_{R0} \in \{-10^{-5}, -10^{-6}\}$, $n \in \{1, 2\}$.

3.8.2 nDGP

The brane-bending mode obeys the Vainshtein equation [22, 37]; in code units

$$\tilde{\nabla}^2 \varphi + \frac{1}{12 \Omega_{rc} \beta(a) a^4} [(\tilde{\nabla}^2 \varphi)^2 - (\tilde{\nabla}_i \tilde{\nabla}_j \varphi)^2] = \frac{\Omega_m a}{\beta(a)} (\tilde{\rho} - 1), \quad (3.4)$$

with $\beta = 1 + 2Hr_c(1 + \dot{H}/3H^2)$ and $F_5 = -\frac{1}{2} \tilde{\nabla} \varphi$, so that the unscreened limit is $G_{\text{eff}}/G = 1 + 1/(3\beta)$ [44].

3.8.3 Symmetron

The dimensionless field $\chi = \varphi/\varphi_\star$ obeys $\tilde{\nabla}^2 \chi = \frac{a^2}{2\tilde{\chi}^2} [\tilde{\rho} (a_{\text{ssb}}/a)^3 - 1] \chi + \frac{a^2}{2\tilde{\chi}^2} \chi^3$ [10, 13], identical to eq. (68) of Brax et al. [6], with force $F_5 = -6\Omega_m \beta^2 (\lambda/B)^2 (a^2/a_{\text{ssb}}^3) \chi \tilde{\nabla} \chi$, whose unscreened limit is the canonical $F_5/F_N = 2\beta^2 \tilde{\chi}^2$. Because $\chi = 0$ is an exact fixed point of the homogeneous equation, the broken-phase VEV $\bar{\chi}(a) = \sqrt{1 - (a_{\text{ssb}}/a)^3}$ is seeded into uninitialised cells at every solve (this also covers restarts and newly refined cells).

3.8.4 Environmentally damped dilaton

The original $r = 3/2$ dilaton of Brax et al. [5], with $A(\varphi) = 1 + \frac{A_2}{2}\varphi^2/M_{\text{Pl}}^2$, $m^2(a) = 3A_2H^2(a)$ and $\beta(a) = \beta_0 a^{3\Omega_m}$ [6], is solved in $\chi = \varphi/M_{\text{Pl}}$:

$$\tilde{\nabla}^2 \chi = \frac{3\Omega_m A_2 B_2}{a} (\tilde{\rho} \chi - \bar{\chi}) + a^2 B_2 [v_\varphi(\chi) - v_\varphi(\bar{\chi})], \quad (3.5)$$

with $v_\varphi(\chi) = -3\Omega_m \beta_0 (A_2 \chi / \beta_0)^{1-3/s}$, $s = 3\Omega_m$, $B_2 = (H_0 B / c)^2$ and $F_5 = -(c/H_0 B)^2 a^2 A_2 \chi \tilde{\nabla} \chi$. The exponent $1-3/s < 0$ makes the Newton Jacobian negative definite. The background is an exact fixed point of the discrete operator, the linear response matches $2\beta(a)^2 k^2 / (k^2 + m^2 a^2)$ to five digits, and a $\delta = 10^5$ top-hat drives $\chi \rightarrow 0.03 \bar{\chi}$ (Damour–Polyakov screening). User parameters are simply β_0 and the present force range $\lambda = 2998 \xi h^{-1} \text{Mpc}$ (then $A_2 = 1/3\xi^2$).

3.8.5 Tracker cubic Galileon

On the tracker, $H\dot{\varphi} = \xi H_0^2 M_{\text{Pl}}$ with $c_2 = -6c_3 \xi$, the model is parameter-free: $\xi = \sqrt{6(1 - \Omega_m)}$, $E^2(a) = \frac{1}{2}[\Omega_m a^{-3} + \sqrt{\Omega_m^2 a^{-6} + 4(1 - \Omega_m)}]$ (wired through the background dispatch, so the modified expansion and growth are automatic; $w_0 \simeq -1.18$), and the perturbation coefficients of Barreira et al. [4] reduce to $\beta_1 = \frac{\xi}{3}[2\dot{H}/H^2 - 1 + (1 - \Omega_m)/E^4]$, $\beta_2 = 2E^2 \beta_1 / \xi^2$. The field equation is the nDGP template with coefficient $1/(3\beta_1 a^4)$ and source $\Omega_m a \delta / \beta_2$; the Poisson back-reaction term integrates to a potential proportional to the field, so the entire fifth force is one gradient, $F_5 = +\frac{\xi}{6E^2} \tilde{\nabla} u$. The unscreened response matches the analytic $-\xi/(9\beta_2 E^2)$ to five digits at $a = 0.25$ –1: $G_{\text{eff}}/G(a=1) = 1.844$, decaying as $1/E^4$ into the past. Since $G_{\text{eff}}/G - 1 < 10^{-3}$ for $z \gtrsim 2.5$, the solver is skipped at early times.

3.8.6 QUMOND

The quasi-linear formulation of MOND [28] defines a curl-free field via $\nabla^2 \Phi = \nabla \cdot [\nu(|\nabla \Phi_N|/a_0) \nabla \Phi_N]$. Both an algebraic force-correction mode and the full phantom-density mode (compute $\rho_{\text{ph}} = \nabla \cdot [(\nu - 1) \nabla \Phi_N] / 4\pi G$, re-solve Poisson) are implemented, with the simple and standard interpolation pairs (verified mutually conjugate to 2×10^{-16}), an external-field option, and a_0 converted to code units through the exact supercomoving acceleration factor (so the MOND scale is correctly constant in proper units at all redshifts). An AQUAL fixed-point mode is also available [cf. 7, 25].

3.9 Numerical solution of the scalar equations

All five scalar solvers share one machinery: nonlinear Newton–Gauss–Seidel relaxation with true three-dimensional red–black colouring (cell parity $i+j+k$), Neumann closure of the stencil at coarse–fine boundaries, one-sided force gradients at level edges, and MPI ghost exchange each sweep. On the fully refined domain level the long-wavelength modes — which plain relaxation cannot converge in $\mathcal{O}(10)$ sweeps — are solved spectrally: a Newton–Helmholtz stage $(\tilde{\nabla}^2 - \bar{m}^2) \delta u = -\mathcal{R}$ with the level-averaged Jacobian mass for $f(R)$ /symmetron/dilaton, and a Winther et al. [44]-style operator splitting for the Vainshtein-type models (solve the cell-local quadratic for $\tilde{\nabla}^2 \varphi$, then invert the Laplacian by FFT). In a controlled test the operator-split stage reached a lower residual in five iterations than 2×10^4 damped relaxation sweeps. Two MPI subtleties deserve mention as a warning to implementers: every rank must enter the per-level ALLREDUCE of the convergence test and the final force ghost-exchange *even when it owns no grids on that level*; early returns on empty ranks deadlock or abort the run at the first sparsely refined level.

Table 2. End-to-end suite: simulated vs. predicted $P/P_{\Lambda\text{CDM}}$ ($k = 0.05\text{--}0.12\ h/\text{Mpc}$, $z \simeq 0$).

Model	Simulation	Forward model	Ratio
Quintessence (RP $\alpha=1$)	0.832	0.817	1.018
k-essence ($X_0/M^4=0.5005$)	0.875	0.881	0.993
Horndeski ($\mu_0=0.2$)	1.036	1.052	0.985
Coupled DE ($\beta=0.1$)	0.966	1.005	0.961
Galileon (tracker)	1.148	1.208	0.950
Dilaton ($\beta_0=0.5$, $\lambda=20$)	1.039	1.071	0.970

3.10 Verification

Verification proceeded on three levels.

Equation level. Every model was re-derived from its original reference (unit conversions included) and the code compared term by term; this stage found and fixed, among others, a wrong-sign DE-perturbation closure, a tachyonic (non-convergent) $f(R)$ operator, a double-counted nDGP coupling, a symmetron force normalisation missing the box-unit conversion, and a zero-pinned symmetron field.

Solver level. The discretised operators were exercised in isolation against analytic linear responses; the accuracies are listed in Table 1.

End-to-end level. A controlled suite of 64^3 , $100\ h^{-1}\text{Mpc}$ dark-matter-only simulations was run from shared Zel’dovich initial conditions ($z = 49$, CAMB spectrum) for ΛCDM , quintessence, k-essence, Horndeski, coupled DE (plus per-effect isolation runs), the tracker Galileon and the dilaton. The measured $P_{\text{model}}/P_{\Lambda\text{CDM}}$ in the band $k = 0.05\text{--}0.12\ h\ \text{Mpc}^{-1}$ at $z \simeq 0$ was compared against an *initial-conditions-consistent* linear forward model, which (i) rescales the initial displacements by each model’s own velocity conversion factor — RAMSES converts the shared velocity files with the model’s v_{fact} , an effect reaching $(1.077)^2$ for k-essence and $(1.109)^2$ for coupled DE — and (ii) propagates the exact (δ, θ) initial vector through each model’s linear system. Results:

All models agree to 2–5 per cent; the Galileon and dilaton deficits carry the expected sign, since Vainshtein and Damour–Polyakov screening remove part of the fifth-force boost already at quasi-linear wavenumbers [cf. 4]. Each coupled-DE ingredient (mass evolution, friction, G_{eff}) was additionally isolated with toggle runs and verified at the ~ 1 per cent level. The forward model, the IC generator and all unit-test drivers ship with the code (`patch/cuRamses/aux/`).

4 Fuzzy Dark Matter

Ultralight scalar dark matter with $m_a \sim 10^{-22}\ \text{eV}$ [15] is evolved as a classical field ψ obeying the Schrödinger–Poisson system. The integrator uses drift–kick–drift operator splitting per AMR level: the kinetic half-steps apply $\exp(-i\hbar k^2 \Delta t / 4a^2 m_a)$ spectrally on the base level (FFT) and an explicit finite-difference Laplacian on refined levels, while the potential kick applies $\exp(-im_a \Phi \Delta t / \hbar)$ in real space; a fourth-order Yoshida composition is optional (`fdm_split_order=4`). Refinement follows the de Broglie wavelength (a minimum of `fdm_nrefine_dB` cells per λ_{dB} , with a density floor), reproducing the solitonic cores and interference fringes characteristic of FDM haloes [36]. A hybrid mode (`fdm_use_hjm`) evolves a Hamilton–Jacobi–Madelung fluid on coarse levels and switches to the wave solver above

`fdm_first_wave_level`, and `fdm_hybrid` allows mixed FDM + particle (stars/sinks) runs. The wavefunction is stored as $(\psi_{\text{re}}, \psi_{\text{im}})$ on the AMR mesh with $|\psi|^2$ the total DM density, and dedicated light-cone and output modules are provided. A detailed description and validation of the FDM solver is presented in a companion paper.

5 Evaporating Primordial Black Holes

The PBH module models dark matter that is partly composed of Hawking-evaporating primordial black holes [8, 12]. Rather than tracking per-particle masses, it uses one global mass function: all DM particles share the mixed-mass factor

$$w(a) = 1 - f_{\text{PBH}} + f_{\text{PBH}} g(a), \quad (5.1)$$

where $g(a)$ is the surviving PBH mass fraction from a precomputed evaporation table and f_{PBH} (`pbh_fraction`) the initial PBH share of the dark matter. Particle masses are rescaled exactly by table arithmetic — there is no per-particle state and no evaporation CFL constraint. The energy released between steps follows the cumulative heating kernel $\tilde{Q}(a) = \int q g dt$; it is either deposited as local heat in the gas (`pbh_energy_sink='local_heat'`, optionally amplified by a linear-response boost factor) or removed from the simulation (`'removed'`, appropriate for free-streaming decay products). Restart provenance (table checksum, normalisation epoch) is persisted so that a run cannot silently continue with a different evaporation model. The table generator supports arbitrary monochromatic or extended PBH mass functions.

6 Conclusions

We have assembled, in one AMR code, the four families of non-standard cosmological physics most often invoked beyond Λ CDM — self-interacting dark matter, a comprehensive dark-energy and modified-gravity battery, fuzzy dark matter, and evaporating primordial black holes — behind uniform namelist switches and on shared solver infrastructure. Beyond the implementations themselves, we highlight the verification methodology: every model is checked at the equation level against its original reference, at the solver level against analytic linear responses, and end-to-end against an initial-conditions-consistent forward model, with per-effect isolation runs where a model bundles several couplings. We commend this three-level protocol to implementers of non-standard physics in any simulation code: in our experience each level catches defects invisible to the others, ranging from wrong-sign perturbation closures to factor-of-two integrator bookkeeping and MPI collective mismatches that only manifest on sparsely refined levels.

All modules recover Λ CDM bit-wise when disabled. The namelist interfaces, verification scripts and the simulation-suite tooling are distributed with the code.

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Data Availability

The code, namelist examples, verification drivers and the 64³ test-suite configuration are available from the cuRAMSES repository. Simulation outputs will be shared on reasonable request.

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